Seaborn Full Course — Beginner to Advanced

Python | Bangla Explanation | Practical Examples | Interview Preparation

1. What is Seaborn?

Seaborn হলো Python-এর একটি statistical data visualization library।
এটি Matplotlib-এর ওপর তৈরি এবং সুন্দর, informative ও statistical charts তৈরি করতে সাহায্য করে।

Why Seaborn?

- Statistical visualization সহজ করে।
- Pandas DataFrame-এর সঙ্গে সহজে কাজ করে।
- সুন্দর default themes ও color palettes।
- Distribution analysis।
- Relationship analysis।
- Categorical data visualization।
- Regression visualization।
- Heatmap ও correlation analysis।

Matplotlib vs Seaborn

Matplotlib	Seaborn
Low-level plotting library	High-level statistical library
More customization control	Easier statistical charts
Manual styling বেশি	Attractive default styles
General visualization	Statistical visualization
NumPy arrays ও অন্যান্য data	Pandas DataFrame integration

2. Installation

```
pip install seaborn matplotlib pandas numpy
```

3. Import Libraries

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

Seaborn version:

```
print(sns.__version__)
```

4. Seaborn Built-in Datasets

Seaborn-এর কিছু sample dataset আছে।

```
import seaborn as sns

print(sns.get_dataset_names())
```

Dataset load:

```
df = sns.load_dataset("tips")

print(df.head())
```

Popular datasets:

Dataset	Description
tips	Restaurant bills and tips
iris	Flower measurements
titanic	Passenger information
penguins	Penguin measurements
flights	Monthly flight passengers

diamonds	Diamond information
exercise	Exercise-related data
mpg	Vehicle fuel economy

Note: `load_dataset()` সাধারণত online dataset repository থেকে data download করে। Internet না থাকলে আগে থেকে downloaded dataset ব্যবহার করতে হবে।

5. Understanding Dataset

```
df = sns.load_dataset("tips")

print(df.head())
print(df.tail())
print(df.shape)
print(df.columns)
print(df.info())
print(df.describe())
print(df.isnull().sum())
```

6. Seaborn Plotting Functions

Seaborn-এর plotting functions কয়েকটি category-তে ভাগ করা যায়:

1. Relational plots
2. Distribution plots
3. Categorical plots
4. Regression plots
5. Matrix plots
6. Multi-plot grids

7. Scatter Plot

দুটি numerical variable-এর relationship দেখায়।

```
df = sns.load_dataset("tips")

sns.scatterplot(
    data=df,
    x="total_bill",
    y="tip"
)

plt.title("Total Bill vs Tip")
plt.show()
```

Add Hue

`hue` দিয়ে category অনুযায়ী আলাদা color করা যায়।

```
sns.scatterplot(
    data=df,
    x="total_bill",
    y="tip",
    hue="sex"
)

plt.show()
```

Add Style and Size

```
sns.scatterplot(
    data=df,
    x="total_bill",
    y="tip",
    hue="sex",
    style="time",
    size="size"
)

plt.show()
```

Important parameters:

Parameter	Purpose
data	Dataset
x	X-axis variable
y	Y-axis variable
hue	Color grouping
style	Marker style
size	Marker size
palette	Color palette
alpha	Transparency

8. Line Plot

Time-series বা continuous data-এর trend দেখায়।

```
df = sns.load_dataset("flights")

sns.lineplot(
    data=df,
    x="year",
    y="passengers"
)

plt.title("Flight Passengers")
plt.show()
```

Multiple Lines

```
df = sns.load_dataset("fmri")

sns.lineplot(
    data=df,
    x="timepoint",
    y="signal",
    hue="region",
    style="event"
)

plt.show()
```

Disable Confidence Interval

নতুন Seaborn version-এ:

```
sns.lineplot(
    data=df,
    x="timepoint",
    y="signal",
    errorbar=None
)

plt.show()
```

পুরোনো version-এ `ci=None` ব্যবহৃত হতো।

9. Distribution Plot

Numerical data-এর distribution বুঝতে ব্যবহার হয়।

9.1 Histogram

```
df = sns.load_dataset("tips")

sns.histplot(
    data=df,
    x="total_bill"
)
```

```
plt.show()
```

Customize Histogram

```
sns.histplot(
    data=df,
    x="total_bill",
    bins=20,
    color="skyblue",
    edgecolor="black"
)

plt.show()
```

KDE Curve

```
sns.histplot(
    data=df,
    x="total_bill",
    kde=True
)

plt.show()
```

9.2 KDE Plot

KDE = Kernel Density Estimation।

এটি distribution-এর smooth density curve দেখায়।

```
sns.kdeplot(
    data=df,
    x="total_bill"
)

plt.show()
```

Multiple KDE

```
sns.kdeplot(
    data=df,
    x="total_bill",
    hue="sex",
    fill=True,
    alpha=0.4
)

plt.show()
```

9.3 ECDF Plot

ECDF = Empirical Cumulative Distribution Function।

এটি দেখায় নির্দিষ্ট value-এর সমান বা কম observation-এর cumulative proportion কত।

```
sns.ecdfplot(
    data=df,
    x="total_bill"
```

```
)

plt.show()
```

9.4 Rug Plot

প্রতিটি observation-এর অবস্থান axis বরাবর ছোট mark দিয়ে দেখায়।

```
sns.rugplot(
    data=df,
    x="total_bill"
)

plt.show()
```

10. Displot

`displot()` হলো figure-level distribution plotting function।

```
sns.displot(
    data=df,
    x="total_bill",
    kde=True
)

plt.show()
```

Compare Categories

```
sns.displot(
    data=df,
    x="total_bill",
    hue="sex",
    kind="kde"
)

plt.show()
```

Facet by Category

```
sns.displot(
    data=df,
    x="total_bill",
    col="time",
    kde=True
)

plt.show()
```

Important distinction:

- `histplot()` → Axes-level।
- `kdeplot()` → Axes-level।
- `ecdfplot()` → Axes-level।
- `displot()` → Figure-level।

11. Box Plot

Distribution, median, quartiles এবং potential outliers দেখায়।

```
sns.boxplot(
    data=df,
    x="day",
    y="total_bill"
)

plt.show()
```

Horizontal Box Plot

```
sns.boxplot(
    data=df,
    x="total_bill"
)

plt.show()
```

Compare Categories

```
sns.boxplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex"
)

plt.show()
```

Box plot concepts:

- Minimum (whisker boundary)
- Q1
- Median
- Q3
- Maximum (whisker boundary)
- Potential outliers

IQR:

```
IQR = Q3 - Q1
```

Common outlier boundaries:

```
Lower = Q1 - 1.5 * IQR
Upper = Q3 + 1.5 * IQR
```

12. Violin Plot

Box plot-এর মতো distribution দেখায়, তবে density shape-ও দেখায়।

```
sns.violinplot(
    data=df,
    x="day",
    y="total_bill"
)
```

```
plt.show()
```

Split by Category

```
sns.violinplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex",
    split=True
)

plt.show()
```

`split=True` ব্যবহার করতে সাধারণত hue-তে দুটি category থাকা প্রয়োজন।

13. Strip Plot

Categorical variable-এর প্রতিটি observation point আকারে দেখায়।

```
sns.stripplot(
    data=df,
    x="day",
    y="total_bill"
)

plt.show()
```

Add Jitter

```
sns.stripplot(
    data=df,
    x="day",
    y="total_bill",
    jitter=True
)

plt.show()
```

Jitter overlapping points আলাদা করে দেখতে সাহায্য করে।

14. Swarm Plot

Strip plot-এর মতো, তবে points overlap কমানোর জন্য automatic positioning করে।

```
sns.swarmplot(
    data=df,
    x="day",
    y="total_bill"
)

plt.show()
```

Large dataset-এ swarm plot computationally expensive হতে পারে।

15. Bar Plot

Categorical groups-এর একটি numerical estimate দেখায়।

```
sns.barplot(
    data=df,
    x="day",
    y="total_bill"
)

plt.show()
```

Default হিসেবে Seaborn mean estimate করে এবং uncertainty interval দেখায়।

Change Estimator

```
sns.barplot(
    data=df,
    x="day",
    y="total_bill",
    estimator="sum",
    errorbar=None
)

plt.show()
```

Custom estimator:

```
import numpy as np

sns.barplot(
    data=df,
    x="day",
    y="total_bill",
    estimator=np.median
)

plt.show()
```

Hue

```
sns.barplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex"
)

plt.show()
```

16. Count Plot

প্রতিটি category-তে কতটি observation আছে তা দেখায়।

```
sns.countplot(
    data=df,
    x="day"
)

plt.show()
```

Hue

```
sns.countplot(
    data=df,
    x="day",
    hue="sex"
)

plt.show()
```

Difference:

- `barplot()` → Numerical values-এর estimate দেখায়।
- `countplot()` → Category-এর observation count দেখায়।

17. Point Plot

Categorical groups-এর estimate এবং uncertainty interval point ও line দিয়ে দেখায়।

```
sns.pointplot(
    data=df,
    x="day",
    y="total_bill"
)

plt.show()
```

Hue

```
sns.pointplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex"
)

plt.show()
```

18. Categorical Plot

`catplot()` হলো figure-level categorical plotting function।
এটি একাধিক categorical plot তৈরি করতে পারে।

```
sns.catplot(
    data=df,
    x="day",
    y="total_bill",
    kind="box"
)

plt.show()
```

Available kinds:

- strip
- swarm
- box

- violin
- boxen
- point
- bar
- count

Example:

```
sns.catplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex",
    kind="violin",
    col="time"
)

plt.show()
```

19. Regression Plot

দুটি numerical variable-এর relationship এবং fitted regression line দেখায়।

19.1 Regplot

```
sns.regplot(
    data=df,
    x="total_bill",
    y="tip"
)

plt.show()
```

Customize

```
sns.regplot(
    data=df,
    x="total_bill",
    y="tip",
    ci=95,
    scatter_kws={"alpha": 0.5},
    line_kws={"color": "red"}
)

plt.show()
```

19.2 Residual Plot

Regression model-এর residual distribution ও pattern পরীক্ষা করতে ব্যবহার হয়।

```
sns.residplot(
    data=df,
    x="total_bill",
    y="tip"
)
```

```
plt.show()
```

19.3 Lmplot

`lmplot()` হলো figure-level regression function।

```
sns.lmplot(
    data=df,
    x="total_bill",
    y="tip",
    hue="sex"
)

plt.show()
```

Facet Regression

```
sns.lmplot(
    data=df,
    x="total_bill",
    y="tip",
    col="time",
    hue="sex"
)

plt.show()
```

20. Heatmap

Heatmap matrix-এর numerical values-কে colors দিয়ে visualize করে।

```
df = sns.load_dataset("flights")

pivot = df.pivot(
    index="month",
    columns="year",
    values="passengers"
)

sns.heatmap(pivot)

plt.show()
```

Customize Heatmap

```
sns.heatmap(
    pivot,
    cmap="viridis",
    annot=True,
    fmt="d",
    linewidths=0.5
)

plt.show()
```

Important parameters:

Parameter	Purpose
data	Matrix data
annot	Show values
fmt	Number formatting
cmap	Color map
linewidths	Cell border width
cbar	Show colorbar
vmin	Minimum color value
vmax	Maximum color value
center	Center value

21. Correlation Matrix

Correlation দুটি numerical variable-এর linear relationship-এর direction ও strength পরিমাপ করে।

```
df = sns.load_dataset("penguins")

numeric_df = df.select_dtypes(include="number")

correlation = numeric_df.corr()

print(correlation)
```

Correlation Heatmap

```
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    center=0
)
```

```
plt.title("Correlation Matrix")

plt.show()
```

Correlation interpretation:

Value	Meaning
+1	Perfect positive correlation
0	No linear correlation
-1	Perfect negative correlation

Important: Correlation does not prove causation।

22. Pair Plot

একটি dataset-এর numerical variables-এর pairwise relationships একসঙ্গে দেখায়।

```
df = sns.load_dataset("iris")

sns.pairplot(df)

plt.show()
```

Hue

```
sns.pairplot(
    df,
    hue="species"
)

plt.show()
```

Select Variables

```
sns.pairplot(
    df,
    vars=[
        "sepal_length",
        "sepal_width",
        "petal_length",
        "petal_width"
    ],
    hue="species"
)
```

```
plt.show()
```

23. Joint Plot

দুটি numerical variables-এর joint distribution এবং marginal distributions দেখায়।

```
df = sns.load_dataset("tips")

sns.jointplot(
    data=df,
    x="total_bill",
    y="tip"
)

plt.show()
```

KDE Joint Plot

```
sns.jointplot(
    data=df,
    x="total_bill",
    y="tip",
    kind="kde"
)

plt.show()
```

Available kinds:

- scatter
- kde
- hist
- hex
- reg
- resid

24. PairGrid

`PairGrid` দিয়ে pairwise plots-এর প্রতিটি অংশ আলাদাভাবে customize করা যায়।

```
df = sns.load_dataset("iris")

g = sns.PairGrid(
    df,
    hue="species"
)

g.map_diag(sns.histplot)
g.map_offdiag(sns.scatterplot)

g.add_legend()

plt.show()
```

25. JointGrid

`JointGrid` দিয়ে joint এবং marginal plots আলাদাভাবে customize করা যায়।

```
df = sns.load_dataset("tips")

g = sns.JointGrid(
    data=df,
    x="total_bill",
    y="tip"
)

g.plot_joint(sns.scatterplot)
g.plot_marginals(sns.histplot)

plt.show()
```

26. FacetGrid

একটি dataset-কে category অনুযায়ী multiple subplots-এ ভাগ করে।

```
df = sns.load_dataset("tips")

g = sns.FacetGrid(
    df,
    col="time",
    row="sex"
)

g.map_dataframe(
    sns.scatterplot,
    x="total_bill",
    y="tip"
)

plt.show()
```

Add Hue

```
g = sns.FacetGrid(
    df,
    col="day",
    hue="sex"
)

g.map_dataframe(
    sns.scatterplot,
    x="total_bill",
    y="tip"
)

g.add_legend()

plt.show()
```

27. Styling and Themes

Seaborn সুন্দর default chart styles দেয়।

Set Theme

```
sns.set_theme()
```

Available Styles

```
sns.set_style("darkgrid")
```

Other styles:

```
sns.set_style("whitegrid")
sns.set_style("dark")
sns.set_style("white")
sns.set_style("ticks")
```

Change Context

```
sns.set_context("notebook")
```

Other contexts:

- paper
- notebook
- talk
- poster

Example:

```
sns.set_theme(
    style="whitegrid",
    context="notebook",
    palette="deep"
)
```

28. Color Palettes

Color palettes charts-এর visual appearance উন্নত করে।

Common Palettes

```
sns.color_palette("deep")
sns.color_palette("muted")
sns.color_palette("bright")
sns.color_palette("pastel")
sns.color_palette("dark")
sns.color_palette("colorblind")
```

Use Palette

```
sns.barplot(
    data=df,
    x="day",
    y="total_bill",
    hue="sex",
    palette="Set2"
)
```

```
plt.show()
```

Display Palette

```
sns.palplot(
    sns.color_palette("viridis", 8)
)

plt.show()
```

29. Customizing Charts with Matplotlib

Seaborn Matplotlib-এর ওপর তৈরি, তাই Matplotlib-এর customization ব্যবহার করা যায়।

```
fig, ax = plt.subplots(figsize=(10, 5))

sns.scatterplot(
    data=df,
    x="total_bill",
    y="tip",
    hue="sex",
    ax=ax
)

ax.set_title("Bill vs Tip")
ax.set_xlabel("Total Bill")
ax.set_ylabel("Tip")

ax.grid(True)

plt.tight_layout()
plt.show()
```

30. Figure-level vs Axes-level Functions

এটি Seaborn-এর গুরুত্বপূর্ণ concept।

Axes-level Functions

এগুলো একটি Matplotlib Axes-এ plot তৈরি করে।

Examples:

- scatterplot()
- lineplot()
- histplot()
- kdeplot()
- boxplot()
- violinplot()
- barplot()
- countplot()
- regplot()

Example:

```
fig, ax = plt.subplots()
```

```
sns.scatterplot(
    data=df,
    x="total_bill",
    y="tip",
    ax=ax
)

plt.show()
```

Figure-level Functions

এগুলো নিজের Figure এবং layout পরিচালনা করে।

Examples:

- relplot()
- displot()
- catplot()
- lmplot()
- pairplot()
- jointplot()

Example:

```
sns.relplot(
    data=df,
    x="total_bill",
    y="tip",
    hue="sex",
    kind="scatter"
)

plt.show()
```

Main Difference

Axes-level	**Figure-level**
Single Axes	Figure and layout
Works with `ax=`	Usually uses `height`, `aspect`, `col`, `row`
Easy Matplotlib integration	Easy faceting

More manual layout control	Automatic multi-plot layout

31. Handling Missing Values

Plot করার আগে missing values পরীক্ষা করা উচিত।

```
df = sns.load_dataset("titanic")

print(df.isnull().sum())
```

Drop Missing Values

```
clean_df = df.dropna()
```

Fill Missing Values

```
df["age"] = df["age"].fillna(
    df["age"].median()
)
```

Plot Clean Data

```
sns.histplot(
    data=df,
    x="age",
    kde=True
)

plt.show()
```

32. Seaborn with Pandas

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "department": ["IT", "HR", "IT", "Sales", "HR"],
    "salary": [50000, 35000, 60000, 45000, 40000],
    "experience": [2, 3, 5, 4, 2]
})

sns.barplot(
    data=df,
    x="department",
    y="salary"
)

plt.show()
```

Grouped Visualization

```
sns.scatterplot(
    data=df,
    x="experience",
    y="salary",
    hue="department"
)

plt.show()
```

33. Seaborn with NumPy

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

data = rng.normal(
    loc=50,
    scale=10,
    size=1000
)

sns.histplot(
    data=data,
    kde=True
)

plt.show()
```

34. Advanced Customization

Remove Spines

```
sns.despine()
```

Rotate X Labels

```
plt.xticks(rotation=45)
```

Set Axis Limits

```
plt.xlim(0, 50)
plt.ylim(0, 20)
```

Figure Size

```
plt.figure(figsize=(10, 6))
```

Tight Layout

```
plt.tight_layout()
```

Save Figure

```
plt.savefig(
    "seaborn_chart.png",
    dpi=300,
    bbox_inches="tight"
)
```

35. Real-World Project 1 — E-commerce Sales Analysis

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "category": [
        "Electronics", "Fashion", "Food",
        "Electronics", "Fashion", "Food"
    ],
    "sales": [5000, 3000, 2000, 7000, 4000, 2500],
    "profit": [1000, 700, 400, 1500, 900, 500]
})

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.barplot(
    data=df,
    x="category",
    y="sales",
    ax=axes[0]
)

axes[0].set_title("Sales by Category")

sns.scatterplot(
    data=df,
    x="sales",
    y="profit",
    hue="category",
    ax=axes[1]
)

axes[1].set_title("Sales vs Profit")

plt.tight_layout()
plt.show()
```

36. Real-World Project 2 — Student Performance Analysis

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "student": ["A", "B", "C", "D", "E"],
```

```
    "math": [80, 70, 90, 60, 85],
    "english": [75, 80, 85, 65, 90],
    "science": [90, 75, 95, 70, 80]
})

numeric_df = df[
    ["math", "english", "science"]
]

correlation = numeric_df.corr()

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    ax=axes[0]
)

axes[0].set_title("Subject Correlation")

long_df = df.melt(
    id_vars="student",
    var_name="subject",
    value_name="marks"
)

sns.boxplot(
    data=long_df,
    x="subject",
    y="marks",
    ax=axes[1]
)

axes[1].set_title("Marks Distribution")

plt.tight_layout()
plt.show()
```

37. Real-World Project 3 — Titanic Data Analysis

```
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset("titanic")

fig, axes = plt.subplots(2, 2, figsize=(12, 9))

# Survival count
sns.countplot(
    data=df,
    x="survived",
    ax=axes[0, 0]
```

```
)

axes[0, 0].set_title("Survival Count")

# Survival by gender
sns.countplot(
    data=df,
    x="sex",
    hue="survived",
    ax=axes[0, 1]
)

axes[0, 1].set_title("Survival by Gender")

# Age distribution
sns.histplot(
    data=df,
    x="age",
    kde=True,
    ax=axes[1, 0]
)

axes[1, 0].set_title("Age Distribution")

# Fare by class
sns.boxplot(
    data=df,
    x="class",
    y="fare",
    ax=axes[1, 1]
)

axes[1, 1].set_title("Fare by Class")

plt.tight_layout()
plt.show()
```

38. Common Mistakes

1. Numeric data-এর জায়গায় categorical variable ব্যবহার করা।
2. Missing values check না করা।
3. `barplot()` এবং `countplot()`-এর পার্থক্য না বোঝা।
4. Correlation-কে causation মনে করা।
5. অনেক বেশি colors ব্যবহার করা।
6. Large dataset-এ swarm plot ব্যবহার করে performance সমস্যা তৈরি করা।
7. Figure-level function-এ `ax=` ব্যবহার করার চেষ্টা করা।
8. Plot-এর axis label না দেওয়া।
9. Distribution বুঝতে শুধু mean-এর ওপর নির্ভর করা।
10. Chart-এর uncertainty interval ভুলভাবে interpret করা।

39. Interview Questions and Answers

Beginner Level

1. What is Seaborn?

Seaborn হলো Python-এর statistical data visualization library।

2. Is Seaborn built on Matplotlib?

হ্যাঁ, Seaborn Matplotlib-এর ওপর তৈরি।

3. How do you import Seaborn?

```
import seaborn as sns
```

4. What is hue?

`hue` দিয়ে category অনুযায়ী data-কে বিভিন্ন color-এ ভাগ করা যায়।

5. What is palette?

Palette chart-এর color scheme নির্ধারণ করে।

6. What is a histogram?

Numerical data-এর frequency distribution দেখায়।

Intermediate Level

7. Difference between barplot and countplot?

- barplot: Numerical data-এর estimate।
- countplot: Category-এর observation count।

8. Difference between boxplot and violinplot?

Boxplot quartiles, median এবং potential outliers দেখায়। Violinplot এর সঙ্গে density shape-ও দেখায়।

9. What is pairplot?

একটি dataset-এর numerical variables-এর pairwise relationships দেখায়।

10. What is heatmap?

Matrix-এর numerical values color দিয়ে visualize করে।

11. What is KDE?

Kernel Density Estimation; distribution-এর smooth density estimate।

12. What is FacetGrid?

Category অনুযায়ী dataset-কে multiple plots-এ ভাগ করার tool।

Advanced Level

13. Difference between relplot and scatterplot?

`scatterplot()` Axes-level; `relplot()` Figure-level এবং faceting support করে।

14. Difference between displot and histplot?

`histplot()` Axes-level; `displot()` Figure-level।

15. What is regression plot?

Data points-এর সঙ্গে fitted regression relationship দেখায়।

16. What is correlation?

দুটি variable-এর linear relationship-এর strength ও direction পরিমাপ।

17. What is PairGrid?

Pairwise visualization-এর প্রতিটি subplot customize করার grid object।

18. What is JointGrid?

Joint distribution ও marginal distribution customize করার grid object।

19. What is estimator in barplot?

প্রতিটি category-এর numerical values থেকে summary statistic বের করে; default হলো mean।

20. What is the difference between figure-level and axes-level functions?

Figure-level functions নিজের figure/layout তৈরি করে; axes-level functions একটি নির্দিষ্ট Axes-এ plot তৈরি করে।

40. Practice Problems

Beginner

1. Load the tips dataset.
2. Create a scatter plot of total_bill and tip.
3. Create a histogram of total_bill.
4. Create a countplot of day.
5. Create a barplot of average tip by day.
6. Create a boxplot of total_bill by day.
7. Create a violinplot of total_bill by time.
8. Add hue="sex" to a scatterplot.
9. Change the chart color palette.
10. Add titles and axis labels.

Intermediate

1. Create a pairplot of iris dataset.
2. Create a heatmap of correlation.
3. Create a KDE plot for total_bill.
4. Create a jointplot of total_bill and tip.
5. Create a regression plot.
6. Create a FacetGrid using day.
7. Create a grouped barplot.
8. Compare male and female distributions.
9. Create a boxplot with hue.
10. Visualize missing data before cleaning.

Advanced

1. Create a complete Titanic EDA dashboard.
2. Analyze correlation between numeric variables.
3. Compare multiple categories using FacetGrid.
4. Build a sales analytics dashboard.
5. Analyze student performance using heatmaps.
6. Visualize outliers using boxplots.
7. Create a regression analysis report.
8. Analyze time-series data.
9. Build reusable plotting functions.
10. Create a complete EDA report using Pandas, Seaborn and Matplotlib.

41. Complete Seaborn Learning Roadmap

Level 1 — Beginner

- Introduction
- Installation
- Built-in datasets
- Scatterplot
- Lineplot
- Histplot

- KDE plot
- ECDF plot
- Barplot
- Countplot

Level 2 — Intermediate

- Boxplot
- Violinplot
- Stripplot
- Swarmplot
- Pointplot
- Catplot
- Displot
- Heatmap
- Pairplot
- Jointplot

Level 3 — Advanced

- Regplot
- Lmplot
- Residplot
- FacetGrid
- PairGrid
- JointGrid
- Figure-level vs Axes-level
- Color palettes
- Themes
- Advanced customization

Level 4 — Professional

- Exploratory Data Analysis
- Statistical visualization
- Correlation analysis
- Outlier detection
- Feature relationship analysis
- Automated visualization
- Dashboard projects
- Data storytelling

42. Recommended Study Routine

Daily Time	Task
30 minutes	Learn concepts

30 minutes	Write code
30 minutes	Practice charts
30 minutes	Analyze datasets

43. Important Functions Cheat Sheet

Function	Purpose
`sns.scatterplot()`	Relationship
`sns.lineplot()`	Trend
`sns.histplot()`	Frequency distribution
`sns.kdeplot()`	Density distribution
`sns.ecdfplot()`	Cumulative distribution
`sns.boxplot()`	Quartiles and outliers
`sns.violinplot()`	Distribution density
`sns.stripplot()`	Individual observations
`sns.swarmplot(`	Non-overlappin

`)`	g observations
`sns.barplot()`	Statistical estimate
`sns.countplot()`	Category count
`sns.pointplot()`	Point estimates
`sns.regplot()`	Regression
`sns.lmplot()`	Figure-level regression
`sns.heatmap()`	Matrix visualization
`sns.pairplot()`	Pairwise relationships
`sns.jointplot()`	Joint distribution
`sns.catplot()`	Figure-level categorical plots
`sns.displot()`	Figure-level distribution
`sns.relplot()`	Figure-level relational plots

`sns.FacetGrid()`	Faceted plots
`sns.PairGrid()`	Custom pairwise plots
`sns.JointGrid()`	Custom joint plots

Final Learning Goal

Seaborn ভালোভাবে শেখার পর তুমি পারবে:

1. Pandas DataFrame থেকে statistical charts তৈরি করতে।
2. Data distribution analyze করতে।
3. Outliers identify করতে।
4. Correlation analyze করতে।
5. Variables-এর relationship বুঝতে।
6. Regression visualization করতে।
7. Exploratory Data Analysis (EDA) করতে।
8. Professional data analysis reports তৈরি করতে।

Recommended next step: Seaborn শেখার পর একটি real-world EDA project তৈরি করো। শুধু chart তৈরি নয়, chart থেকে data সম্পর্কে meaningful insights বের করার practice করো।